

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

From the Industrial Revolution to 2024 | 1750 – 2024

Data Source: Global Carbon Budget 2025 · NASA Vitals Signs · Our World in Data

For Policy Analysts & Climate Negotiators

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million (ppm)

Mauna Loa Observatory, Feb 2026

Source: NOAA / NASA

Pre-Industrial Baseline

~280 ppm

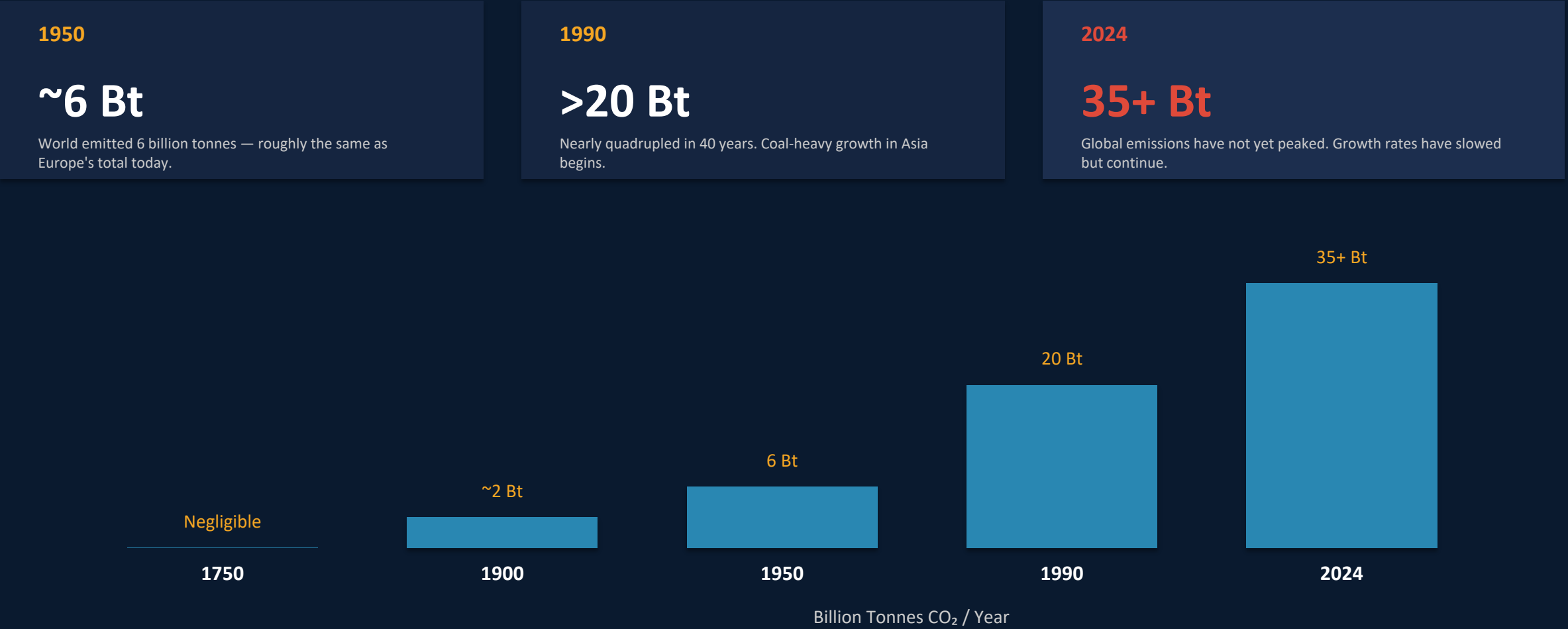
Before the Industrial Revolution, atmospheric CO₂ was stable at ~280 ppm — a level sustained for thousands of years.

The Acceleration

- Current CO₂ is 1.5× the natural ice-age increase — which took millennia
- The rise from 280 → 429 ppm occurred in ~2 centuries, driven by coal, oil, and gas combustion
- Ground-based and satellite measurements confirm sharp post-1950 acceleration

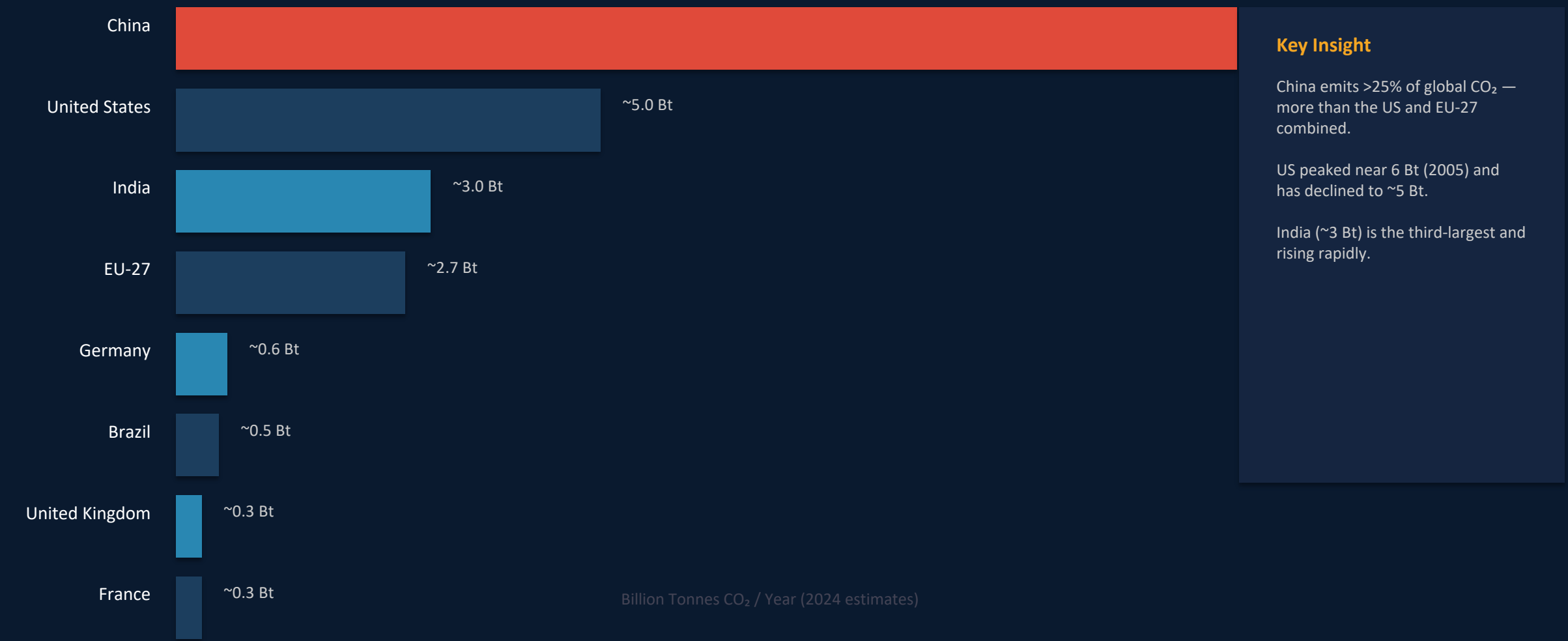
Note: figure_1.jpg provides the per capita emissions line chart over time.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



Reference: figure_2.jpg — Global emissions trajectory line chart

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



Reference: figure_3.jpg — Annual CO₂ emissions by country

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900



2024



1900

Europe + US: >90%

Asia: <5%

Africa: <3%

2024

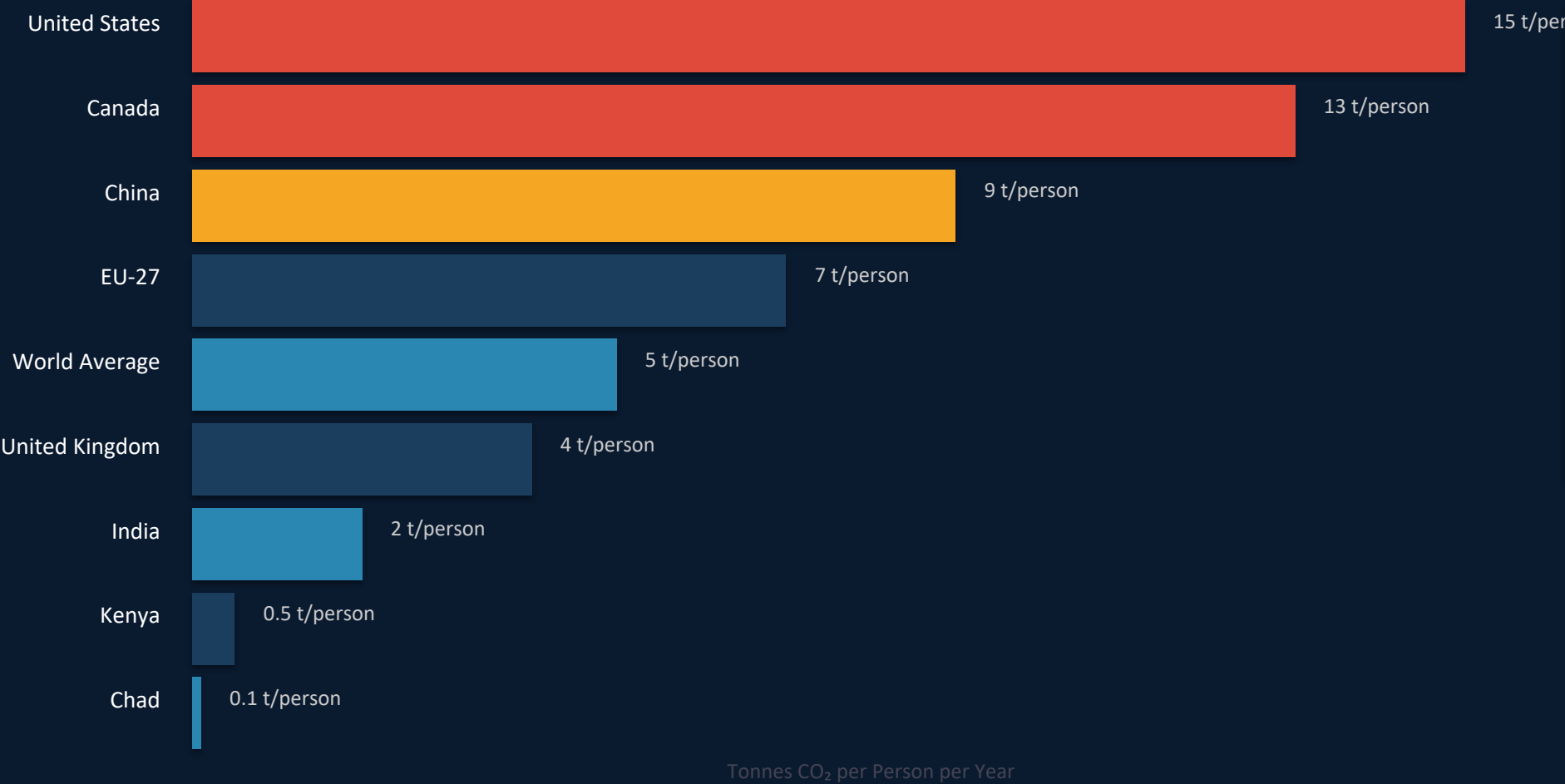
Asia: ~50%

US + Europe: <33%

Africa: 3–4% | South America: 3–4%

Note: This dramatic shift reflects industrialization of Asia (especially China and India) over the past 75 years. Africa and South America together emit less than international aviation and shipping.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



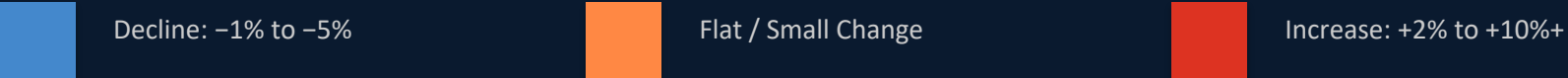
150×

disparity

An American emits as much CO₂ in under 2 days as the average person in Chad or Niger emits in an entire year.

Reference: figure_1.jpg — Per capita CO₂ emissions line chart over time

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



United States & Europe

- United States: -1% to -3%
- Germany: -2% to -4%
- United Kingdom: -1% to -3%
- France: -1% to -2%
- Overall EU: declining

Russia, Africa & SE Asia

- Russia: +2% to +5%
- Southeast Asia: +3% to +7%
- Parts of Africa: +2% to +10%
- Some African nations >+10%
- Driven by coal & oil demand

South America & Other

- Mixed signals across region
- Some countries: >+10%
- Others: declining
- High variability
- Limited data coverage

Reference: figure_4.jpg — Annual percentage change map in CO₂ emissions, 2024

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on energy sources.

Country	Nuclear+Renewables	Fossil Fuel Share	Per Capita CO ₂
France	~75%	~25%	~4 t
United Kingdom	~50%	~50%	~4 t
Germany	~30%	~70%	~8 t
Netherlands	~20%	~80%	~8.5 t

France vs. Germany

- France: ~75% nuclear + renewables → only ~25% fossil fuels → ~4 t/capita
- Germany: ~30% nuclear + renewables → ~70% coal & gas → ~8 t/capita
- Both are advanced economies with similar GDP per capita
- Energy mix choice = policy choice → directly controls per capita emissions

Policy Implication

Prosperity alone does not determine per capita emissions — the energy mix does.

Countries that invest in nuclear and renewables (France, UK, Sweden) have per capita CO₂ 2–3× lower than those reliant on coal and gas (Germany, Netherlands, Poland).

This has direct implications for climate negotiations: emissions per capita reflects energy policy, not just wealth.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

01

Global emissions exceed 35 Bt and have not yet peaked

Despite slowing growth rates, annual CO₂ emissions continue to rise. The world emitted ~6 Bt in 1950, ~20 Bt in 1990, and over 35 Bt in 2024.

02

China alone emits >25% of global total (~12.5 Bt)

More than double the United States (~5 Bt). India (~3 Bt) is the third-largest and rising rapidly, posing a significant challenge to global decarbonization efforts.

03

Per capita inequalities remain extreme — a 150× gap

The US (~15 t/person) and Australia (~15 t) emit 150× more per capita than Chad or Niger (~0.1 t). An American emits in under 2 days what a Chadian emits in a year.

04

US and Europe are declining, but historical responsibility remains large

The US peaked near 6 Bt around 2005 and has fallen to ~5 Bt. Europe is also declining. However, cumulative historical emissions mean developed nations carry major responsibility for atmospheric stock.

05

Energy mix — not just prosperity — determines per capita emissions

France (~4 t/capita) vs. Germany (~8 t/capita) proves this. Policy and infrastructure choices around nuclear, renewables, and fossil fuels directly shape per capita outcomes — a critical negotiating point.